MILITARY SPECIFICATION

SPONGE, SURGICAL, GAUZE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tamperproof sealed trays containing sterile radiopaque sponges.

1.2 Classification. The sponges shall be of one type and of the following sizes: (see 6.3).

Type	Size
I	4 x 8 inches, 800s: 80 trays in a box
II	4 x 4 inches, 1280s: 128 trays in a box

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a aprt of the specification to the extent specified herein:

SPECIFICATIONS

Federal

UU-P-31	Paper, General Specifications and Methods of Testing.
CCC-G-101	Gauze, Absorbent.

FSC 6510

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

U.S. PHARMACOPEIAL CONVENTION, INC.
Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY

T4100-S-68 Weight per Unit Area (Basis Weight of Substance) of Paper Paperboard

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York, N. Y. 10017).

3. REQUIREMENTS

3.1 Material.

3.1.1 Gauze. Gauze shall be type VII gauze in accordance with Fed. Spec. CCC-G-101 of the issue in effect on date of invitation of bids.

3.1.2 Precount feature. The precount feature shall be either a colorfast string or a laminated band of polypropylene and tissue ¼ inch wide minimum. The string shall be tested for colorfastness in accordance with 4.3.3. Results of the test shall be good. There shall be no appreciable change from the original color of the string. Staining of the cloth shall be reason for rejection of the string. The bands shall be tested as specified in 4.3.4 and shall show no signs of decomposition or parting at the seal.

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3.1.3 X-Ray detectable insert. The x-ray detectable insert shall be either in the form of a monofilament or rectangular tab. Monofilament or tab shall be fabricated of either textile or plastic material, and shall be impregnated with an inert, nontoxic substance which is opaque to x-rays. If a monofilament is used, it shall be approximately equal to the length of the gauze in the warp direction, and not less than 0.381 mm. thick prior to use in the sponge. If a tab is used, it shall measure not less than 0.5 square inches, and the length shall be a minimum of four times the width. The insert shall appear prominent and conspicuous when covered by at least 5/8 inches of aluminum when tested as specified in 4.3.2.

3.1.4 Paperboard for trays. Paperboard for trays shall be solid-bleached sulfate, 140 pounds per ream (24 x 36 - 500) minimum, when tested as specified in par. 4.3.5.

3.1.5 Paper for covers. Paper for covers shall be white, bleached, sterilizable kraft, basis weight 45 pounds per ream (24 x 36 - 500) minimum, when tested as specified in par. 4.3.6.

3.1.6 Plastic for covers. Plastic for covers shall be white sheet material of high density spun bonded polyethylene fibers. Sheet shall be 8 mils thick.

3.1.7 Plastic for tray body. Plastic for tray body shall be high impact styrene, 0.030 inches thick before forming.

3.1.7.1 Plastic for tray body, reinforced type construction. When the construction is such that the plastic is used for reinforcement, the plastic shall be polypropylene.

3.2 Construction.

3.2.1 Gauze pads. Both sizes of gauze pads shall be made of 12 plies of type VII gauze 20 by 12 mesh (3.1.1), shall be substantially even in shape, size and thickness, and shall be folded in such a manner that they may be opened to approximately the sizes in Table I without exposing any raw edges or loose threads. The area of the piece of gauze shall be not less than shown in Table I. The radiopaque tab or monofilament used in the 4 by 8 size shall be so anchored between the layers of gauze that it will pass the test in 4.3.7 (Not fall out). The inserted tab or monofilament shall also pass the live steam sterilization test (4.3.8) without bleeding, becoming tacky, or showing any other signs of deterioration.

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Table I - Gauze Areas

Size	Minus * Tolerance	Number of plies	Open Size (Approximate)	Area of gauze (Minimum)
4 x 4 inch	1/4	12	4 x 12	192
4 x 8 inch	1/2	12	4 x 24	380

*Maximum minus tolerance allowed in either dimension or in the sum of two dimensions.

3.2.2 Tray. Construction of tray bodies shall be either of the constructions shown in the figures. Material may be either styrene (3.1.7) or a combination of paper (3.1.4) and polypropylene. (3.1.7.1)

3.2.3 Tray and gauze pad assembly. The complete assembly shall consist of either of the following:

NSN 6510-00-116-1235 - Ten 4 by 8 radiopaque sponges in a tray of suitable size (3.2.2). Sponges shall be bonded with a precount feature (3.1.2). Cover shall be either paper (3.1.5) or plastic (3.1.6). Assembly shall be sealed and sterile.

NSN 6510-00-116-1311 Ten 4 by 4 sponges in a tray of suitable size (3.2.2). Cover shall be either paper (3.1.5) or plastic (3.1.6). Assembly shall be sealed and sterile.

Construction shall be in accordance with the figures for the applicable stock numbers, design and materials. When the trays are tested for opening, they shall pass the parting test (4.3.9).

3.2.4 Labeling of Covers. Covers shall have the following labeling, or equivalent, in easily legible letters:

Name, address and registered trademark of the manufacturer.
Peel back instructions in one corner.
Number, size and ply of sponges.
If sponge contains a radiopaque element, identification of the sponge as such.
The legend "Sterility Guaranteed Unless Package is Damaged or Opened." or similar legend.

3.3 Sterility. The sponges shall be sterile when tested with a biological indicator as specified in 4.4.

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3.3.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification
b. National stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore, strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa., 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.3.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.2. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.4 Workmanship. Workmanship shall be first class throughout. The finished sponges, components, and trays shall be free from defects which will detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the paper and the plastic, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in Table I and III (See 4.2.1 and 4.2.2).

4.2.1 Sampling for Examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table II. Unit of product for examination purposes shall be the tray.

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Table II - Sampling for Examination

Defects	Inspection Level	A. Q. L.
Major A	S-4	1.0
Total A	S-4	4.0
Major B	S-2	1.0
Total B	S-2	4.0
Minor A	S-4	2.5

4.2.2 Examination. The sponges shall be examined for defects including, but not limited to those listed in Table III.

Table III - Classification of Defects

Major A

101 Tray and lid not free from cuts, holes and tears.
102 Tray capable of being resealed.
103 Number of plies and area of gauze of sponges not as specified.

Major B

150 Dimensions not within specified tolerances.

Minor A

201 Labeling or printing on lid not legible and free from defacing marks or scratches.

4.2.3 Sampling for Tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one sponge. Each lot of material used in making the sponges and trays shall be tested at least once.

4.3 Tests.

4.3.1 Gauze. Each lot of gauze used shall be tested in accordance with the latest edition of the United States Pharmacopeia for Absorbent Gauze.

4.3.2 X-Ray Detectable Insert. In testing for opacity, the insert or inserts shall be placed on an unexposed x-ray film and covered with an aluminum ladder ranging in thickness from at least 1/8 to 3/4 inch in steps of 1/8 inch. X-ray exposure shall be approximately 10MAS, 80KVP, to meet the contract requirements, at least 30 inches using M speed screens in a convenient size cassette.

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4.3.3 Test for Colorfastness of Pre-Counted Thread Feature. A thread approximately six feet in length, from each lot of thread shall be wrapped in a cloth two inches by four inches. Cloth shall be bleached but not starched, cotton cloth 80 x 80 print cloth or finer. The wrapped thread shall then be placed in a sterilizer and subjected to saturated steam at a minimum of 240° F for not less than 20 minutes after sterilization conditions have been obtained, (when gauge pressure corresponds to that of saturated steam at the indicated temperature).

4.3.4 Sterilization Capability of Pre-Counted Laminated Band Feature. At least 5 bands, taken at random from each rool of laminated material shall be tested. The bands shall be placed in a sterilizer and subjected to saturated steam at a minimum of 240° F not less than 20 minutes after sterilization conditions have been attained (when gauge pressure corresponds to that of saturated steam at the indicated temperature).

4.3.5 Basis weight, paperboard for trays. Basis weight, paperboard for trays, shall be determined in accordance with TAPPI Method T410OS-68, Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.

4.3.6 Basis weight, paper for tray covers. Basis weight, paper for tray covers, shall be determined in accordance with TAPPI Method 410OS-68, Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.

4.3.7 Anchoring Test. The gauze pad containing the monofilament of tab shall be opened to the open size as indicated in Table I, and shaken vigorously. The pad shall pass the requirement in 3.2.1.

4.3.8 Sterilization Test, radiopaque gauze pads. The radiopaque gauze pads shall be sterilized by steam at a temperature of 250° F for 1 hr. After cooling, the monofilaments or inserts shall not show any deterioration specified in 3.2.1.

4.3.9 Parting test. When the trays are opened as directed, the part shall be such that the sponges shall fall freely from the opening.

4.4.1 Sterility test. The sponge and tray assemblies shall be tested for sterility in accordance with the method specified in the current USP for Purified Cotton, Gauze, Surgical Dressings and related materials using inoculated product, inoculated simulated product, or paper spore strip (carrier). Where carriers are used they shall be placed inside the tray between the sponges. Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in paragraph 4.4.1.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units.

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4.4.1.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C
Dry Heat	B. stearothermophilus, or	55° - 65°C
	B. subtilis var.niger (or globigii)	30° - 35°C
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.1.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of tests units shall be as indicated in 4.4.1 Sterility test and 4.4.1.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.1.1 NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.1.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.1.4 Product testing. Testing for sterility of the product after receipt by the Government shall be without a biological indicator and in accordance with the U.S.P.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package containing eighty trays of 800 sponges, 4 by 8 inch or one hundred twenty eight trays of 1280 sponges 4 by 4 inches, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Tray. Ten sterile sponges of one size shall be packaged in a sealed tamperproof tray as specified in 3.2.3.

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5.1.2.2 Unit package. Eighty trays of 4 by 8 inch sponges or 128 trays of 4 by 4 sponges shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, Type CF, class weather-resistant except that W5c shall not be acceptable. The box shall conform to the special requirements as specified in Table II of PPP-B-636. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.1.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Packing. Units packaged as specified in 5.1.2 or 5.1.3 shall require no further packing.

5.3 Marking.

5.3.1 Tray. Each tray shall be marked as specified in 3.2.4.

5.3.2 Unit package. Each unit package shall be marked in accordance with the requirements of the Federal Food, Drug and Cosmetic Act. In addition, each unit package shall be marked in accordance with MIL-STD-129 as required for exterior containers. Marking shall include the sterility lot (control) number and the method of sterilization (G-Gas, R-Radiation, S-Steam, C-Cold, D-Dry Heat Sterilization) shall be shown following the level of packaging and date of packaging. Example: "A-4/73-G". Final sterilization shall be on or after indicated packaging date.

6. NOTES

6.1 Intended use. The sponges covered by this specification are sterile pre-counted sponges supplied in sealed trays for use in emergency conditions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type, required (see 1.2).

(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock Number	Type	Item Identification
6510-00-116-1285	I	SPONGE, SURGICAL, Gauze, Radiopaque, 4 by 8 inches, 800s: Twelve-ply. Sterile. Ten sponges sealed in a tray with peelable cover; 80 trays in a box.
6510-00-116-1311	II	SPONGE, SURGICAL, Gauze, 4 by 4 inches, 1280s: Twelve-ply. Sterile. Ten sponges sealed in a tray with peelable cover; 128 trays in a box.

6.4 This specification does not include all types, classes, grades or sizes of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

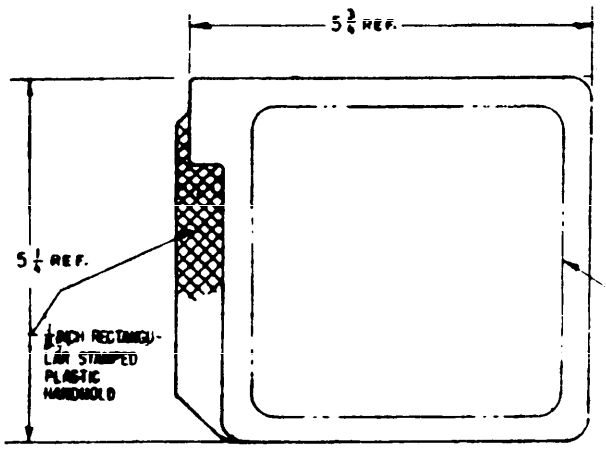
Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6510-0588

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

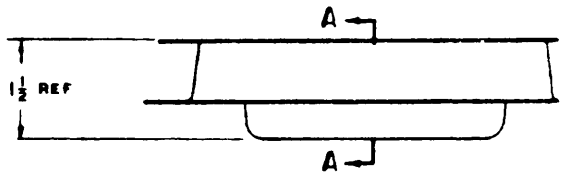
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TRAY COVER - HIGH DENSITY SPUN BONDED POLYETHYLENE 8 MILS THICK

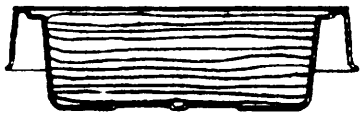
PHANTOM LINES REPRESENT SEAL AREA - ADHESIVE TO BE APPLIED FROM PHANTOM LINE TO OUTER EDGE OF THE TRAY. PRINTING NOT TO BE LOCATED IN SEAL AREA.

TOP



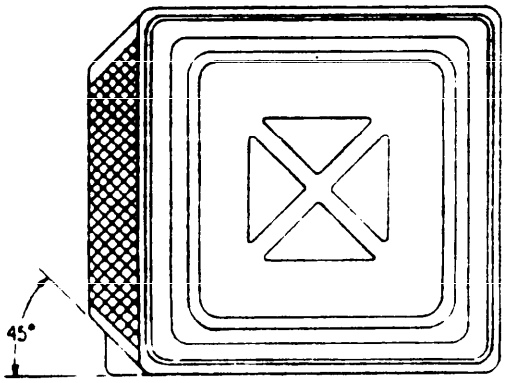
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TRAY MAT'L - POLYSTYRENE 0.030 THICK MIN BEFORE FORMING



SEC-A-A

10-4 BY 4, 12 PLY SPONGES IN SEAL TRAY



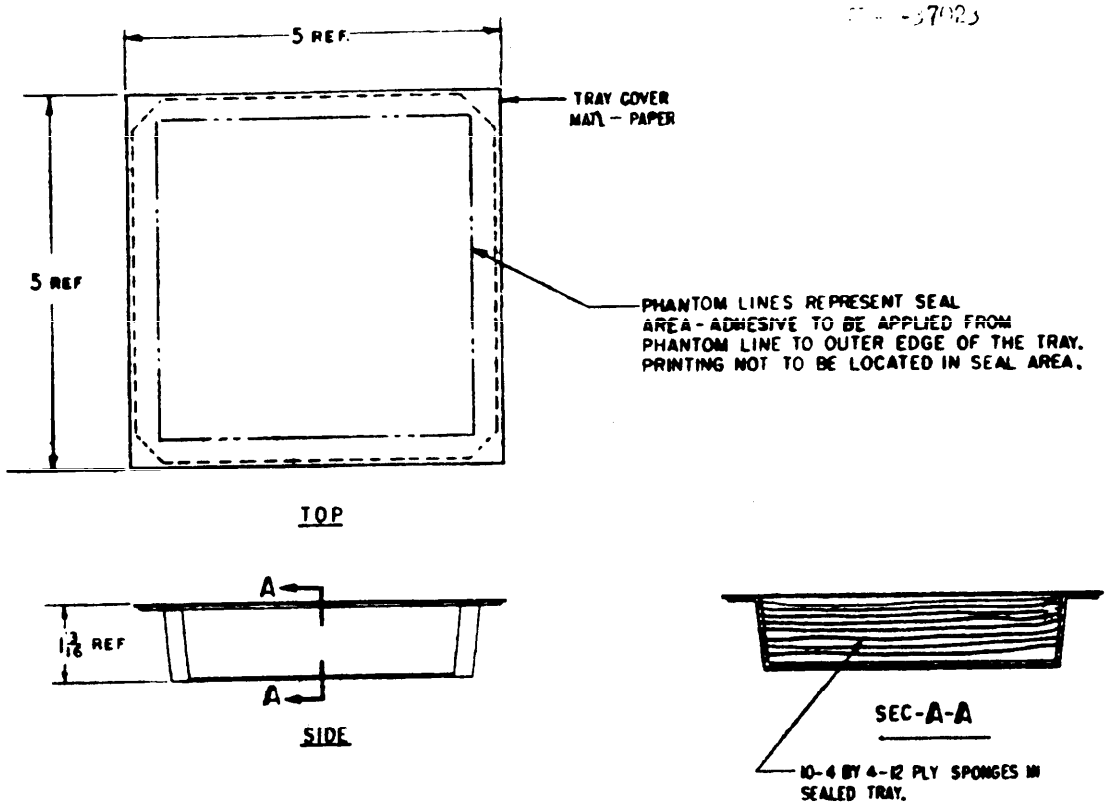
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NSN 6510 00 116-1311
SPONGE AND TRAY ASSEMBLY
TRAY TYPE - ALL POLYSTYRENE

FIGURE 1

DWG 23100 F
SHEET 13

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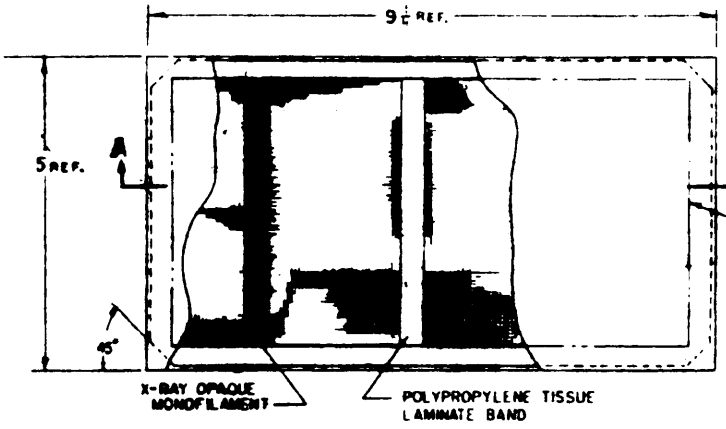


NSN 6510-00-116-1311
SPONGE & TRAY ASSEMBLY
TRAY TYPE - POLYPROPYLENE AND PAPERBOARD

FIGURE 2

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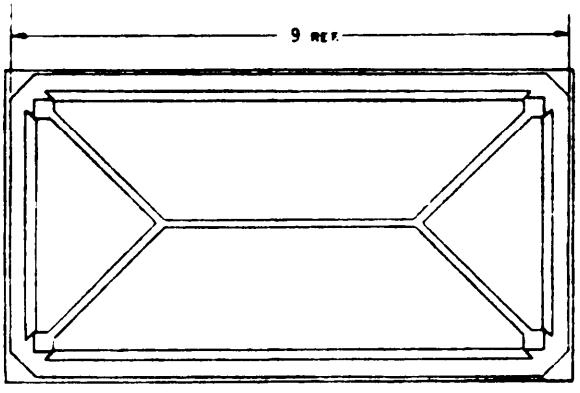
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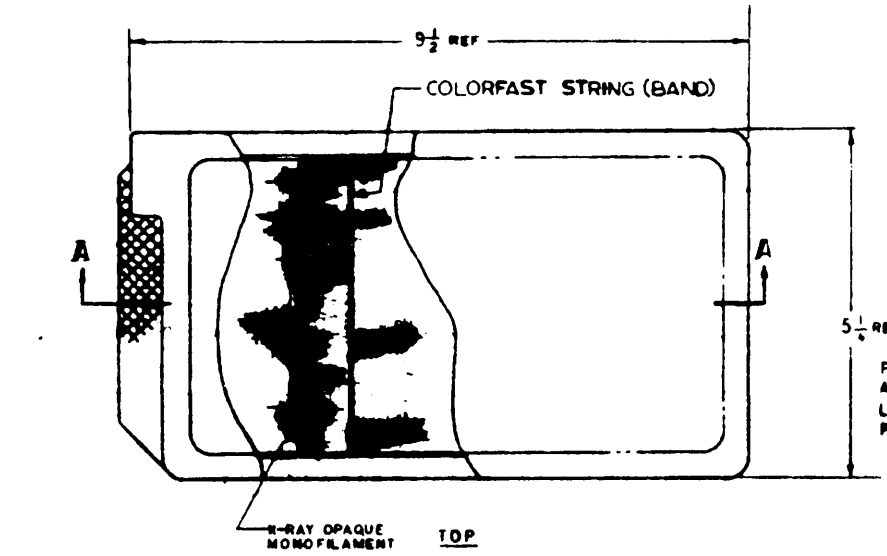


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NSN 6510-00-116-1285
SPONGE AND TRAY ASSEMBLY
TRAY TYPE - POLYPROPYLENE AND PAPERBOARD

FIGURE 3

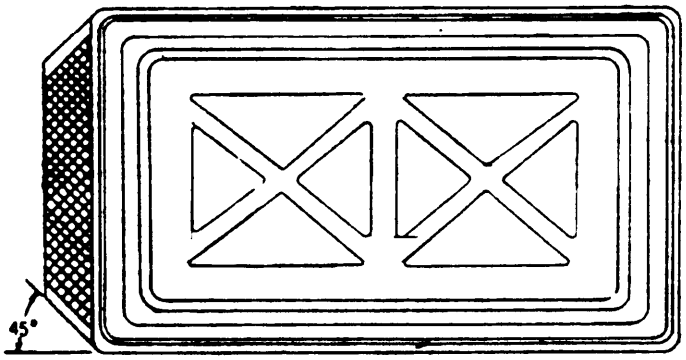
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PHANTOM LINE REPRESENTS SEAL AREA ADHESIVE TO BE APPLIED FROM PHANTOM LINE TO OUTER EDGE OF THE TRAY. PRINTING TO BE LOCATED IN SEAL AREA.



SEC-A-A



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NSN 6510-00-116-1285
SPONGE AND TRAY ASSEMBLY
TRAY TYPE-ALL POLYSTYRENE

FIGURE 4

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

C 16061

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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DEFENSE SUPPLY AGENCY
DoD - 304

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PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
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2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

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Check the source to verify that this is the current version before use.